Divesh Haresh Valecha

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Education

Bachelor of Engineering in Artificial Intelligence & Data Science *2023 - 2027*
Thadomal Shahani Engineering College, Mumbai, India (Average CGPA 9.61)

Higher Secondary Certificate *February 2023*
Smt. Chandibai Himathmal Mansukhani College, Ulhasnagar, India (72.50%)

Work Experience

Deloitte 15th June 2026 – Current
Tech Intern in Audit & Assurance Function
- Independently designed and built a journal-entry (JE) sampling tool as a custom embedded widget on Deloitte's GenW platform, implementing a memory-bounded two-pass CSV pipeline with seeded random sampling for audit-defensible selection; currently being evaluated for firm-wide production rollout.
- Engineered a full-stack audit analytics dashboard that automates spreadsheet ingestion, consolidates manual defaulter-detection checks into one automated pass with visual analysis & auto-generated list, and emails flagged defaulters directly.

Meanttor December 2025 – March 2026
Web Developer Intern
- Designed REST APIs with Django REST Framework and PostgreSQL, implementing JWT-based authentication and role-based access control.
- Built an AI-powered personalized study planner using GPT-4o and Qdrant (RAG pipeline), automating scheduling with Python RQ for asynchronous task execution.

Projects

Snippet Vault – AI-Powered Code Snippet Management Platform May 2025
Tech Stack *- Java, Spring Boot, Spring Security, MySQL, React, REST APIs, Qdrant, OpenAI API*
- Developed a platform to store, manage, and retrieve code snippets with secure user-specific access.
- Designed a reference-based forking system, letting users fork another user's snippet into their own collection without duplicating the underlying data.
- Built a Chrome extension to capture and save snippets directly from the browser, improving accessibility and user experience.
- Implemented AI-powered semantic search and snippet explanation using embeddings and LLMs for enhanced developer productivity.

mygit – Git Implementation Built from Scratch March 2026
Tech Stack *- Java 21, Picocli, Maven, HttpServer, OpenAI API*
- Built a complete Git implementation from scratch in Java 21, within a 24-hour hackathon, implementing SHA-1 content-addressable storage and zlib compression to replicate Git's internal object model.
- Engineered a three-way merge algorithm using LCS-based diffing to handle conflict detection and resolution, along with stash and gitignore support for full workflow parity with real Git.
- Integrated an OpenAI-powered natural language agent, enabling users to query repository state and perform Git operations through conversational commands.

NicheIQ — AI-Powered YouTube Campaign Intelligence Platform April 2026
Tech Stack *- FastAPI, Next.js 14, YouTube Data API, Twilio, APScheduler, Python*
- Automated data collection via YouTube Data API v3, building labelled datasets from 1,000+ videos and fine-tuning a DistilBERT classifier to categorize videos into 20 content niches.
- Engineered a multi-agent workflow orchestrating the full audit pipeline - translation, classification, CPV benchmarking, and report generation - with zero manual intervention.
- Automated report delivery via APScheduler cron jobs pulling Google Ads placement data, dispatched as personalized WhatsApp/Telegram reports through Twilio on a user-defined schedule.

Skills

Programming Languages: Java, Python, C
Backend & Web: Spring Boot, Spring Security, React, REST APIs, Django, Django REST Framework, FastAPI
Databases: MySQL, PostgreSQL, SQL, Supabase, Firebase
Core Concepts: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks
Tools & Technologies: Git, GitHub, Postman, VS Code, Redis, Kafka
AI / Agentic AI: LangChain, RAG Pipelines, Vector Databases
Problem Solving:
- Solved 250+ problems on LeetCode (DSA, problem solving, optimization)

Achievements

Deloitte Hacksplosion 2026 *Top 10 out of 21,000+ participants*
- Led the team behind **ARIA**, an autonomous industry research agent built on Deloitte's proprietary GenW.AI platform, helping clients evaluate a market before entering it.
- Engineered the core pipeline that sourced, reconciled, and synthesized data into a decision-ready market report.

Research Papers

CF-CGAN: A Novel Architecture for Data Augmentation in Low-Data Regimes *March 2026*
Accepted and published at the Third International Conference on Computing Systems and Intelligent Applications
Authors: Jeet Suchdev, Divesh Valecha, Himani Deshpande, Jayesh Ruchandani, Karan Ramrakhiani